

MindBridge AI Wellness Agent

Comprehensive Project Report

Course: Artificial Intelligence (5th Semester)

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Assignment: Assignment #1 - AI Agent Web App

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Project Domain: Mental Wellness & AI Healthcare Support

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1. Executive Summary

MindBridge AI Wellness Agent is an intelligent, multi-feature web application designed to support mental health and wellness through AI-powered conversations. Built as part of the Artificial Intelligence course (5th Semester), this project demonstrates practical implementation of Large Language Model (LLM) APIs integrated with a modern web framework.

The agent operates in the **mental wellness and AI healthcare support domain**, providing four core services: Dream Interpretation, Mood Analysis, Mindfulness Guidance, and Crisis Support. Powered by the Groq LLM API (llama-3.1-8b-instant model), MindBridge delivers empathetic, insightful, and actionable responses to users seeking emotional support.

Metric	Value
Framework	Streamlit
LLM API	Groq (llama-3.1-8b-instant)
Programming Language	Python 3.14
Features	4 Core Modules
Response Format	Structured Markdown
Deployment	Localhost (Development)

2. Project Overview & Objectives

2.1 Problem Statement

Mental health awareness is growing, yet many individuals lack immediate access to emotional support tools. Traditional therapy requires appointments, costs, and stigma barriers. MindBridge addresses this gap by providing an AI-powered wellness companion available 24/7, offering immediate emotional guidance, dream interpretation, mood analysis, and crisis intervention.

2.2 Objectives

- Build a functional AI agent using free LLM APIs (Grog)
- Implement a multi-feature wellness application with distinct modules
- Create an intuitive, aesthetically pleasing user interface
- Ensure ethical AI practices (privacy, non-diagnostic, crisis escalation)
- Demonstrate practical AI agent development skills for academic assessment

2.3 Domain

Primary Domain: Mental Health & Wellness Support

Secondary Domain: Conversational AI / Healthcare AI

Application: AI-powered emotional support, mood tracking, dream interpretation, crisis detection

3. System Architecture & Tech Stack

3.1 Architecture Diagram

- Client Layer:** Web Browser (Chrome, Firefox, Edge)
- Presentation Layer:** Streamlit UI (Python-based web framework)
- Application Layer:** Python business logic, API handlers, session management
- AI Layer:** Groq LLM API (llama-3.1-8b-instant model)
- Configuration Layer:** Environment variables (.env) for API keys

3.2 Technology Stack

Component	Technology	Purpose
Frontend Framework	Streamlit 1.38.0	Interactive web UI with tabs, sliders, forms
Backend Language	Python 3.14	Core application logic
HTTP Client	Requests 2.31.0	API communication with Groq
Environment Config	python-dotenv 1.0.1	Secure API key management
LLM API	Groq API	Fast inference with open-source models
Model Used	llama-3.1-8b-instant	Text generation and interpretation

4. Feature Breakdown

■ Dream Weaver

Analyzes user-described dreams using symbolism, Jungian psychology, creative storytelling, and reflection. Provides structured output with four sections: Symbolism, Psychological Insight, Creative Interpretation, and Reflection.

■ Mood Mirror

Accepts user emotional state descriptions and a 1-10 mood rating. Returns emotion detection, mood breakdown (stress/energy/optimism levels), insight into causes, and three suggested wellness activities.

■ Mindful Mentor

Provides personalized guided meditation, breathing exercises, and mindfulness tips based on user needs. Includes step-by-step exercises, daily mindfulness tips, and reflection questions.

■ Crisis Companion

Offers immediate emotional validation, grounding techniques, and professional resource referrals. Includes emergency services banner and strongly encourages professional help for severe cases.

5. AI Agent Implementation

5.1 System Prompts

Each feature uses a carefully crafted system prompt that defines the agent's role, response format, and tone. Prompts are designed to produce consistent, structured, and empathetic outputs.

Dream Weaver Prompt Structure:

Role: Mystical dream interpreter

Format: Four sections with emojis (🔮, 📖, 🌙, 🌿)

Constraints: 2-3 sentences per section, insightful but not clinical

5.2 Session State Management

Streamlit's session state maintains user data across interactions. The sidebar includes a 'Clear History' button that resets all session variables, ensuring privacy and fresh starts.

5.3 Error Handling

Comprehensive error handling covers: missing API keys, API errors (400, 401, 429), unexpected response formats, and network failures. Users receive clear, actionable error messages instead of raw stack traces.

6. API Integration (Groq LLM)

6.1 Groq API Overview

Groq provides ultra-fast inference for open-source LLMs. The free tier offers generous limits suitable for development and academic projects. API keys are obtained from console.groq.com.

6.2 API Request Structure

Endpoint: POST <https://api.groq.com/openai/v1/chat/completions>

Headers: Authorization (Bearer token), Content-Type: application/json

Payload: model, messages array (system + user), temperature (0.7), max_tokens (600)

6.3 Model Selection Journey

Initially, the project used **llama3-8b-8192**, which was later decommissioned by Groq. This required migration to **llama-3.1-8b-instant**, demonstrating adaptability in working with evolving API ecosystems.

7. User Interface Design

7.1 Design Philosophy

The UI follows a **light, calming aesthetic** with soft gradients (blue-purple-pink), white cards, and vibrant accent colors. The design prioritizes accessibility, readability, and emotional comfort.

7.2 Layout Components

Component	Description
Sidebar	User profile, wellness stats, clear history button
Header	Project title with gradient styling
Stats Bar	Visual metrics (tools count, 24/7 support, privacy)
Tab Navigation	4 clickable tabs with icons for feature switching
Input Areas	Textareas with placeholder hints for user guidance
Mood Slider	1-10 range slider for quantitative mood input
Action Buttons	Color-coded gradient buttons per feature
Result Cards	Styled output containers with left accent borders

8. Testing & Validation

8.1 Test Cases

Test Case	Input	Expected Result	Status
Dream Interpretation	I was flying over an ocean	Structured 4-section response	■ Pass
Mood Analysis	I feel overwhelmed, rating 3	Emotion detection + activities	■ Pass
Mindfulness Guide	Help me sleep better	Guided exercise + tips	■ Pass
Crisis Support	I feel hopeless	Validation + resources + hotline	■ Pass
Empty Input	(blank)	Error: Please enter text	■ Pass
Missing API Key	(no key)	Error: API key not configured	■ Pass
Invalid Model	llama3-8b-8192	API Error 400: Model decommissioned	■ Handled

8.2 Performance Metrics

- Average Response Time:** 2-4 seconds per query
- API Success Rate:** >95% with valid key and model
- UI Load Time:** <1 second on localhost
- Concurrent Users:** Single user (development server)

9. Ethical Considerations

- **Non-Diagnostic:** The agent explicitly states it is not a substitute for professional mental health care.
- **Crisis Escalation:** Crisis Companion includes emergency service numbers and strongly encourages professional help.
- **Privacy:** API keys are stored in .env files (not hardcoded). Session data can be cleared via sidebar button.
- **Transparency:** System prompts are designed to be warm, supportive, and non-clinical. No false medical claims.
- **Bias Awareness:** LLM responses may contain biases. The structured prompt format helps constrain outputs to wellness-focused content.
- **Data Handling:** No user data is permanently stored. Conversations exist only during the active session.

10. Challenges & Solutions

Model Decommissioning

Problem: Groq deprecated llama3-8b-8192, causing API Error 400.

Solution: Migrated to llama-3.1-8b-instant with updated model string.

TemplateNotFound Error

Problem: Flask could not locate index.html due to wrong working directory.

Solution: Switched to Streamlit framework, eliminating template dependency.

API Key Configuration

Problem: .env file not loading properly in some environments.

Solution: Implemented fallback hardcoding for testing; emphasized .env best practices.

Dependency Management

Problem: pip and streamlit commands not recognized in PATH.

Solution: Used python -m pip and python -m streamlit for module execution.

File Path Confusion

Problem: Multiple app.py files in different directories caused execution errors.

Solution: Created clean project folder structure with single entry point.

11. Future Enhancements

- **User Authentication:** Login system to save personal wellness history
- **Database Integration:** SQLite/PostgreSQL for persistent mood tracking over time
- **Voice Input:** Speech-to-text for hands-free dream/mood logging
- **Multi-language Support:** Urdu, Arabic, and other regional languages
- **Wearable Integration:** Connect with fitness trackers for sleep/heart rate correlation
- **Clinician Dashboard:** Supervised care interface for mental health professionals
- **Community Features:** Anonymous peer support forums
- **Mobile App:** React Native or Flutter companion app

12. Conclusion

MindBridge AI Wellness Agent successfully demonstrates the practical application of AI agents in the mental health domain. Through integration of the Groq LLM API with Streamlit, the project delivers four distinct wellness features: Dream Weaver, Mood Mirror, Mindful Mentor, and Crisis Companion.

The project overcame significant technical challenges including model decommissioning, framework migration (from Flask to Streamlit), and environment configuration issues. These experiences provided valuable insights into real-world AI development workflows, API management, and responsive UI design.

With its ethical design principles, structured response formats, and calming user interface, MindBridge represents a meaningful step toward accessible AI-powered mental wellness support. Future iterations will expand functionality while maintaining the core commitment to user privacy and emotional safety.

13. References & Resources

- Groq API Documentation: <https://console.groq.com/docs>
- Streamlit Documentation: <https://docs.streamlit.io>
- Python Requests Library: <https://docs.python-requests.org>
- python-dotenv: <https://saurabh-kumar.com/python-dotenv/>
- LLaMA Model Papers: Meta AI Research
- Jungian Dream Analysis: Carl Gustav Jung, 'Man and His Symbols'
- Mental Health First Aid Guidelines: WHO Mental Health Action Plan 2013-2030

End of Report